Digital Mohalla — Class diagram of the whole system (design step A7 v1.0: iteration 2 v3.0, DA-11): Modules 1, 3, 4, 5, 7, 9 and 10, all three students

Python design classes. Colour = owner: S1 Fatima S2 Areeba S3 Minahil . **Green text** = added or changed in iteration 2; **green border** = class new in iteration 2. Each class names the FRs it realises; a dependency on another module is the interface named in the attribute or operation (module dependencies: component diagram).

accounts (Module 1)

User
id: UUID
display_name: str
verification_level: VerificationLevel
platform_role: PlatformRole
status: AccountStatus
is_synthetic: bool
terms_accepted_at: datetime | None
FR-A-01, FR-A-04 (part), NFR-S-07; no CNIC / address fields

Membership
user_id: UUID
community_id: UUID
role: MembershipRole
status: MembershipStatus
is_moderator_of(community_id): bool
FR-A-09 (seeded), NFR-S-03

«value»
LocalityResult
membership_id: UUID
method: VerificationMethod
outcome: LocalityOutcome # never negative
geofence_signal: GeofenceSignal | None
invitation_id: UUID | None
permission_refused, mock_provider_detected, duplicate_suspected: bool
approved_by: UUID | None # moderator path
checked_at, valid_until: datetime
FR-A-07, FR-A-08, NFR-PR-02 (no coordinates)

Invitation
community_id, inviter_id: UUID
code_hash: bytes # SHA-256 only
expires_at: datetime
FR-A-07, UC-A3 3d (A11; single use)

«service»
GeofenceInvitationVerifier
evaluate(fix, permission_refused, invitation, duplicate_suspected) : LocalityResult # fix used in memory only
FR-A-07, FR-A-08 (A11 decision table)

«interface»
LocalityChecker
outcome_for(user_id, community_id): LocalityOutcome
FR-B-07 (UC-B5 step 4)

StoredResultLocalityChecker
outcome_for(…): LocalityOutcome # latest valid stored result
FR-B-07, FR-A-08; reads results written by the verifier

AuthService
login(identifier, password): Session
current_actor(token): Actor
FR-A-04 (part), NFR-S-01

uses 0..1 / 0..1 · 0..* / 1 · 1 / 0..* · «use»

Notes
• lifecycle_state (where the loan is) and AssessmentResult.assurance_state (what the evidence says) are separate attributes of separate classes (ES-04. HELD_FOR_REVIEW with REVIEW_RECOMMENDED).
• BorrowingTransaction is rebuilt only through apply(); nothing sets lifecycle_state directly (FR-B-09).
• Student 2 implements AssuranceEngine and OutcomeSink; Student 1 implements IndirectTrustEngine, which the AssuranceEngine calls through IIndirectEvidence.
• Module 9 (Student 3) calls BorrowingService.apply_decision() through ILifecycleCommands; it is not drawn here.
• Relationship ends: ◆ composition, ◇ aggregation, dashed open arrow = dependency, dashed hollow triangle = realisation.
• Green = added or changed in iteration 2; the scheduler (JobRunner) calls BorrowingService like any user command.

«use»

lifecycle (Module 3)

«application service»
BorrowingService
- store: EventStore
- validator: TransitionValidator
- engine: AssuranceEngine
- locality: LocalityChecker
- outcomes: OutcomeSink
- notify: INotify
create_request(cmd: CreateRequest): BorrowingTransaction
accept(bt), actor) / decline(id, actor, reason)
cancel(id, actor)
record_handover_checklist(id, actor, report)
confirm_handover(id, actor, agreed)
initiate_return(id, actor)
confirm_return(id, actor, report)
open_dispute(id, actor, reason) / withdraw_dispute(id, actor)
apply_decision(id, decision: ModeratorDecision)
counter_offer(id, lender_period) # 107
respond_counter(id, borrower, accept) # 729 / I08
request_extension(id, borrower, new_due) # I03
respond_extension(id, lender, approve) # I04 / I05
expire(id) / no_show(id) / remind(id, step) # scheduler
escalate_overdue(id) # T24
FR-B-05 to B-07, B-10 to B-18

«aggregate»
BorrowingTransaction
id: UUID # = stream_id
item_id, borrower_id, lender_id: UUID
requested_period: DateTimeRange
lifecycle_state: LifecycleState
hold_reason: HoldReason | None
due_at: datetime | None
stream_version: int
apply(event: LifecycleEvent): None
replay(events): BorrowingTransaction «static»
FR-B-08, FR-B-09, NFR-R-03

«create, load»

«value, immutable»
LifecycleEvent
event_id: int
stream_id: UUID
stream_seq: int
event_type: EventType
schema_version: int
actor_id: UUID | None
occurred_at: datetime
previous_state, new_state: LifecycleState
evidence_refs: dict
authorisation: str
config_version: str | None
payload: dict # ADR-0005
FR-B-09, FR-N-03, NFR-D-01 (frozen schema)

«use» · «ordered» · 1 / 1..* · «stores»

TransitionValidator
- table: TransitionTable
validate(txn, event_type, target, actor): Transition
raises InvalidTransition, NotAuthorised, GuardFailed
FR-B-08, NFR-R-02, NFR-S-03, NFR-S-03

TransitionTable
rows: list[Transition] # 39, from transition_table_v3.csv
lookup(state, event_type, target): Transition | None
FR-B-08, NFR-R-02 (A3 table = test matrix)

«interface»
EventStore
append(stream_id, expected_seq, event): None
raises StaleStream
load(stream_id): list[LifecycleEvent]
FR-B-09, ADR-0002 (unique stream_id + stream_seq)

PostgresEventStore
append(…) # one DB transaction: event + booking
load(…)
FR-M-05, NFR-S-05 (append-only, hash chain)

«value»
Transition
from_state, to_state: LifecycleState
event_type: EventType
allowed_roles: frozenset[ActorRole]
guard: Callable[[Context], bool]
ref. str # T01..T29 (T25a–c), R1..I08
FR-B-08

Booking
transaction_id, item_id: UUID
period: DateTimeRange
status: BookingStatus {HELD, RELEASED}
FR-B-05, NFR-R-01 (EXCLUDE constraint, ADR-0002)

0..1 / 0..*

«value»
ConditionReport
phase: ConditionPhase
reporter_role: PartyRole
checklist: dict
condition_agreed: bool
deposit_received: bool | None
reduced_evidence: bool
FR-B-11, FR-B-13; I-4 record-only deposit

«enumeration»
LifecycleState / EventType / HoldReason
14 states, 24 event types (+3 counter-offer, ADR-0015), 5 hold reasons
Possession: NOT_HANDED_OVER | WITH_BORROWER | RETURNED | LOST
(lists: State_Machine_Design_v3.md)
FR-B-08, FR-B-09, ADR-0003, ADR-0009

«value»
Actor
user_id: UUID | None
role: ActorRole
authorisation: str
NFR-S-02

assurance.indirect (Module 5, S1 part)

TrustEdgeProjector
on_outcomes(outcomes: list[Outcome]): None
FR-T-04 (edges from verified outcomes only, FR-T-03)

«upsert»

TrustEdge
source_id, target_id: UUID
role: PartyRole
metric: OutcomeMetric
weight: float # Beta posterior mean, ADR-0006
evidence_count: int
FR-T-04

1 / 0..* · 1 / 1

TrustGraph
- edges: dict[UUID, list[TrustEdge]]
neighbours(u, role, metric): list[TrustEdge]
FR-T-04

«use»

KHopPathSearch
k: int # from config (initial 3)
path(u, v, role, metric): list[Path] # simple paths, len <= k
FR-T-04, NFR-P-03 (k sweep)

«creates» · «use»

«value»
Path
nodes: list[UUID]
hops: int
q: float
FR-T-04

1..* · «value»
IndirectEvidence
paths: list[Path]
strength_i: float
rho: float # <= 0.6
1_indirect: float
prior_backed: bool
FR-T-04, FR-T-01 (paths in explanation, ES-13)

«returns»

IndirectTrustEngine
gamma, cap: float # config
indirect_evidence(lender, borrower, role, metric, t_v, b): IndirectEvidence
FR-T-04 (realises IIndirectEvidence)

assurance (Module 5, Student 2)

«interface»
IAssess
assess(request: AssessmentRequest): AssessmentResult
- raises EngineTimeout (2,000 ms)
FR-T-01, FR-T-07

«interface»
IOutcomeSink
record_outcome(closing_event_id: int): None {idempotent}
FR-B-15, NFR-R-04

ConfigLoader
- path: str
load(): EngineConfig
NFR-D-01

«value»
EngineConfig
config_version: str
poor_half_life, interval_level
k, gamma, cap, divisors, baseline
signal weights, bands, rounding
thresholds, value_bands, timeout_ms
NFR-D-01 (Assurance_Engine_Config_Draft.json)

«value»
AssessmentRequest
transaction_id: UUID
borrower_id, lender_id: UUID
item_id: UUID
role: Role, metric: Metric
FR-B-07, FR-T-01

«use»

AssuranceFacade
- extractor: OutcomeExtractor
- estimator: BetaPosteriorEstimator
- indirect: IIndirectEvidence
- detector: ManipulationDetector
- policy: DecisionPolicy
- explainer: ExplanationBuilder
- config: EngineConfig
assess(request): AssessmentResult
record_outcome(closing_event_id)
FR-T-01, FR-T-07, FR-B-15, NFR-D-01

OutcomeExtractor
extract(closing_event): list[OutcomeEvent]
eligible(subject, role, metric): list[OutcomeEvent]
FR-T-03 (confirmed outcomes only; disputed excluded)

BetaPosteriorEstimator
estimate(subject, role, metric, events, profile?)
→ TrustEstimate # roles never mixed
posterior_mean(s_w, f_w): float (shared, ADR-0006)
weightage_days, similarity(): float
repeat_discount(events): weights # same pair
FR-T-02, FR-T-03

ExplanationBuilder
build(direct, indirect, risk, decision): list[EvidenceRecord]
rendered(ids) # str
FR-T-08, NFR-U-01 (never one number)

DecisionPolicy
- thresholds: tau_mu, tau_L, n_min, rho_sup
decide(direct, indirect, risk, ctx)
→ DecisionPolicyResult
rules R1–R5, R6_DISPUTE, R6_SPARSE, R7–R9
order R1, R2, R4, R3, … (ADR-0010)
FR-T-06, ADR-0007, ADR-0008, DM-09

«creates» · «returns» · «creates» · «returns» · «creates»

«value»
AssessmentResult
state: AssuranceState
direct: TrustEstimate
indirect: IndirectEvidence
manipulation: ManipulationRiskAssessment
decision: DecisionPolicyResult
explanation: list[EvidenceRecord]
config_version: str
computed_at: datetime
FR-T-01, NFR-D-01 (stored as one assessment row)

«value»
OutcomeEvent
event_id, transaction_id
subject_id, observer_id
role: Role, metric: Metric
success: bool
status: EvidenceStatus
item_category, value_band, duration_days
occurred_at: datetime
FR-T-03 (persisted as outcome)

«value»
TrustEstimate
alpha, beta: float
mean: float
lower_90, upper_90: float
effective_sample_size: float
direct_event_count: int
FR-T-02

«value»
EvidenceRecord
factor: str
value: str
confidence: float
supporting_event_ids: list
FR-T-08

«value»
DecisionPolicyResult
decision: DecisionAction
rule_fired: str
hold_reason: HoldReason | None
explanation facts
escalate: bool (= hold set)
FR-T-06, ADR-0003

«use» · 1 / 1..* · 1 / 0..*

RetaliationCheck
- window_days: int = 14
is_suspected(review, records): bool
FR-T-09, ES-05

ManipulationDetector
- signals: list[SignalCheck]
- weights, bands
- corroboration: 2 signals ≥ 0.5
- S1 to S5 all active (S2, S3, S5 new)
assess(subject, window): ManipulationRiskAssessment
FR-T-05, ADR-0007

«value»
ManipulationRiskAssessment
score: float [0..100]
level: int {1..4}
corroborated: bool
FR-T-05

«value»
TransactionContext
item_value, value_band
is_high_value, medium_value: bool
open_dispute: bool
resolved_disputes_12m: int
verification_ok: bool
request_similarity: float
FR-T-06

«value»
SafeguardRecommendation
type: SafeguardType
parameters: dict
reason: str
FR-T-01

1 / 1..* · 1 / 1..* · 1 / 1..*

«interface»
SignalCheck
name: str
weight: float
evaluate(subject, data): ManipulationSignal
FR-T-05

«value»
ManipulationSignal
name: str
available: bool
value: float [0..1]
weight: float
evidence: list[str]
FR-T-05 (evidence, never accusation)

«enumeration»
AssuranceState, DecisionAction, EvidenceStatus, SafeguardType
AssuranceState: SUFFICIENT_EVIDENCE | REVIEW_RECOMMENDED | INSUFFICIENT_EVIDENCE | ACTIVE_DISPUTE
DecisionAction: PROCEED | PROCEED_WITH_SAFEGUARDS | REQUEST_VERIFICATION | RESTRICT_HIGH_VALUE | ESCALATE
EvidenceStatus: CONFIRMED | DISPUTED | RESOLVED (Assurance DD §5)
SafeguardType: security_deposit | handover_photos | return_date_fixed | …

ReciprocalClustering
S1, weight 0.30
evaluate(…)
FR-T-05, ES-04

ReviewSimilarity
S4, weight 0.20
evaluate(…) {char n-gram TF-IDF}
FR-T-05, ES-04, ADR-0007

RatingVelocity
S2: ratings in 7 d vs 83 d
evaluate(…)
FR-T-05, ADR-0007

NewAccountActivity
S3: age < 30 d, 14 d count
evaluate(…)
FR-T-05, ADR-0007

InvitationLinkage
S5: invite tree, 90 d
evaluate(…)
FR-T-05, ADR-0007

Notes
• AssuranceFacade is the only entry point: Module 3 calls IAssess.assess() once per request (A8); a timeout gives an engine_failure hold, never a default state (FR-T-07).
• Trust and manipulation risk stay separate attributes of AssessmentResult and are never averaged (FR-T-06).
• Three signals (velocity (S2), NewAccountActivity (S3) and InvitationLinkage (S5)) are the iteration 2 SignalCheck realisations (green).
• BetaPosteriorEstimator.posterior_mean is also the trust-edge weight (ADR-0006).
• Relationship ends: ◆ composition, ◇ aggregation, dashed open arrow = dependency, dashed hollow triangle = realisation.

accounts.membership (Module 1)

«application service»
RegistrationService
register(email, password, display_name, terms_accepted): User
same generic answer if the e-mail exists (FR-A-01)
FR-A-01, NFR-S-01 (UC-A1)

«application service»
MembershipService
- locality: GeofenceInvitationVerifier
- invitations: InvitationService
- notify: INotify
communities(query): list[CommunitySummary]
request_join(user, community, Locality#qui, code)
- Membership # OPEN + VERIFIED: approved at once
queue(moderator): list[MembershipRequest]
decide(moderator, membership, kind, reason): MembershipDecision
reverse(moderator, decision, reason)
FR-A-06, A-09, A-10, A-11 (UC-A3, UC-A4)

«service»
InvitationService
issue(member, community): str # code shown once
redeem(code, community): Invitation
FR-A-07, UC-A3 3a

«entity»
MembershipDecision
membership_id, decided_by: UUID
kind: MembershipDecisionKind
reason: str
reverses_decision_id: UUID | None
decided_at: datetime
FR-A-10, FR-A-11 (append-only)

«enumeration»
MembershipDecisionKind
APPROVED, REJECTED,
INFO_REQUESTED, REVERSED

lifecycle.listings and scheduling (Module 3)

«value»
CounterOffer
proposed_period: DateTimeRange
offered_at: datetime
open: bool
from_events(events): CounterOffer | None # replay
FR-B-10, ADR-0015

«application service»
ListingService
- embeddings: EmbeddingJob
create(lender, draft): Item
update(lender, id, changes): Item
publish(id) / paused(id) / expired()
FR-B-01, FR-B-18 (UC-B1)

«entity»
Item
id, owner_id, community_id, category_id: UUID
title: str
value_band: ValueBand
deposit_amount: Decimal | None # record-only
status: ItemStatus
description: str
condition: ItemCondition
usage_terms, handover_method: str
search_terms: str # e.g. Roman Urdu names
expires_at, updated_at: datetime
can_publish(): bool # description + condition
FR-B-01 (schema v2.4)

«application service»
JobRunner
- jobs: list[Job]
- runs: JobRunStore
register(job: Job): None
run_due_jobs(now, as_of=None): JobReport
advisory lock; claim, act, finish
FR-B-18, NFR-R-05, ADR-0012 (UC-B13)

«interface»
Job
name: str
due(now): list[DueItem]
run(item): None
FR-B-18

NoShowJob
due: ACCEPTED past grace
FR-B-12

OverdueJob
due: ACTIVE past limit
FR-B-14

ExpiryJob
due: ASSESSED past window
FR-B-10

«port»
JobRunStore
claim(job, due_key): bool # unique (job, key)
finish(id, status, error)
NFR-R-05 (job_run table)

search.retrieval (Module 4, S1)

«interface»
ISearch
search(member, query, filters, sort): SearchResults
FR-S-01 to S-03 (used by the app through the REST API)

«application service»
SearchService
- embedder: EmbeddingProvider
- ranker: HybridRanker
- log: SearchLogger
search(member, query, filters, sort): SearchResults
retrieve(query, vector | None, filters, limit): list[Candidate]
one SQL query: full-text rank + vector distance;
community, PUBLISHED, not blocked inside it (I-5)
no vector: keyword-only, mode KEYWORD_ONLY (FR-S-02)
FR-S-01, S-02, S-03, ADR-0016 (sequence i)

«value»
SearchFilters
category_id: UUID | None
dates: DateTimeRange | None
max_deposit: Decimal | None
condition: ItemCondition | None
sort: RELEVANCE | EVIDENCE | NEWEST
FR-S-03 (no radius: no coordinates)

«value»
Candidate
item_id: UUID
keyword_rank: float
similarity: float | None
FR-S-01

«value»
SearchResults
query_id: UUID
mode: HYBRID | KEYWORD_ONLY
results: list[RankedResult]
FR-S-01, FR-S-05

assurance.reviews and lender evidence (Module 5)

«value»
RoleMetric
role: BORROWER | LENDER
metric: ON_TIME_RETURN | CONDITION_AGREED | DISPUTE_FREE
Overview §5.2 (never mixed)

«service»
PopulationBaseline
baseline(role, metric): float # fallback 0.5 below 30
refresh(now) # scheduled job
FR-T-04 baseline b, DA-15

«service»
LenderEvidence
- estimator: BetaPosteriorEstimator
- baseline: PopulationBaseline
lender_factors(lender, profile): list[EvidenceRecord]
ranking_value(lender): float # internal only
FR-S-04, FR-S-01; never one number to users

«application service»
ReviewService
- retaliation: RetaliationCheck
submit(actor, txn, ratings, comment): Review
edit(actor, review, changes) # edit window
FR-T-09, FR-B-15 (UC-D1)

«create»

«entity»
Review
transaction_id, reviewer_id, reviewee_id: UUID
ratings: dict, comment: str
retaliation_flag, similarity_flag: bool
set by the service, confirmed by a moderator
FR-T-09 (table since schema v2.3)

search.ranking (Module 4, S2)

«port»
EmbeddingProvider
model: str # chosen by the memory test, ADR-0016
embed(texts): list[Vector] # may raise ModelUnavailable
FR-S-01, FR-S-02 fallback

«service»
EmbeddingJob
due(now): items whose text hash changed
run(item) # item_embedding row
FR-B-03 (retry later), ADR-0012

«service»
HybridRanker
- weights: RankingWeights
- lenders: LenderEvidence
rank(candidates, ctx): list[RankedResult]
reason(result): str # one line, no score
FR-S-01, Algorithm 5 (sequence i)

«value»
RankingWeights
relevance 0.40, lender_evidence 0.30,
availability 0.15, same_community 0.10,
freshness 0.05; config_version
NFR-D-01 (config 1.4.0)

«value»
RankedResult
item_id: UUID
score_parts: dict # stored, not shown
reason: str
FR-S-01, FR-S-05

notifications (Module 7)

«interface»
INotify
notify(user, type, payload, dedupe_key): None
FR-C-02

«application service»
NotificationService
notify(user, type, payload, dedupe_key): None # once per key
inbox(user, unread_only): list[Notification]
mark_read(user, ids)
FR-C-02 (in-app), FR-B-06, FR-B-17, ADR-0013 (UC-G3)

«entity»
Notification
user_id: UUID
type: str
payload: dict # no location
dedupe_key: str
created_at, read_at: datetime
FR-C-02

moderation (Module 9)

«interface»
IModeration
queue(community_id, filters): list[CaseSummary]
case_detail(case_id): CaseDetail
request_information(case_id, party, question, reason)
record_decision(case_id, kind, reason, …)
file_appeal(case_id, party, reason)
decide_appeal(case_id, upheld, reason)
reverse(decision_id, reason)
audit_log(filters): list[AuditEntry]
FR-T-10, FR-T-11, FR-M-05 (REST: /cases, /audit-log; endpoints in DA-10)

«interface»
ICaseIntake
open_or_attach(transaction_id, trigger, reason): EscalationCase
withdraw(transaction_id)
FR-T-10; UC-D6 1b (one live case per transaction)

AuditLogService
record(actor, action, target, outcome, reason?)
query(filter: AuditFilter): list[AuditEntry]
{masked by default; the view is logged}
unmask(entry_id, purpose): AuditEntry
{logged}
verify_chain(): ChainCheck
concentration(moderator_id, window)
FR-M-05, NFR-S-05, NFR-S-03, UC-I6 2–4, 3b–3d

AuditEntry, AuditFilter
AuditEntry: occurred_at, actor_id,
authorisation, action, target_type,
target_id, previous_value, new_value,
outcome SUCCESS | REJECTED, reason,
prev_hash, row_hash
AuditFilter: actor, user, transaction,
case, date range, action, result,
community (own only)
FR-M-05 (audit_log row), UC-I6 2

«use» · «use» · «use»

CaseService
- cases: CaseRepository
- notifier: INotify
- attachments: AttachmentService
- config: timer values
open_or_attach(transaction_id, trigger, reason)
queue(community_id, filters)
case_detail(case_id)
request_information(case_id, party, question, reason): deadline = now + window (v2, C02)
add_statement(case_id, party, text, files) {before the deadline: C03}
expire_information(now): int (v2, C04; idempotent, called by the scheduler)
withdraw(transaction_id)
route_to_admin(case_id)
FR-T-10, FR-T-11 (step 2), FR-B-17, UC-D6 2a, ADR-0012

EvidenceTrailBuilder
- events: IEventQuery
build(case): EvidenceTrail
gaps(trail): list[str]
FR-T-10; UC-D6 1, 1a (trail derived from the event log)

DecisionService
- lifecycle: ILifecycleCommands
- audit: AuditLogService
record_decision(case_id, moderator, kind, reason, safeguards?, resolved_outcome?, possession?): ModeratorDecision
allowed_kinds(case): set[DecisionKind] (ADR-0003)
allowed_possession(previous_state) : set[Possession] (v2, ADR-0009)
FR-T-11, ADR-0003, ADR-0004, ADR-0009, UC-I4 3, 3c, 3d, 4a

«enumeration»
CaseTrigger, CaseStatus, DecisionKind, Possession
CaseTrigger: DISPUTE | MANIPULATION_FLAG | HIGH_VALUE_LOW_CONFIDENCE | LOCALITY_HOLD | ENGINE_FAILURE
CaseStatus: OPEN | AWAITING_INFORMATION | DECIDED | APPEALED | REVERSED | WITHDRAWN
DecisionKind: CLEARED | SAFEGUARDS_IMPOSED | RESTRICTED | DISPUTE_RESOLVED | APPEAL_UPHELD | APPEAL_OVERTURNED | REVERSAL
Possession: NOT_HANDED_OVER | WITH_BORROWER | RETURNED | LOST

«creates» · «creates» · «use» · «use»

1 / 0..* manages

EscalationCase
id, transaction_id: UUID
opened_event_id: int
trigger_type: CaseTrigger
hold_reason | dispute_reason
raised_by: UUID | None
status: CaseStatus
priority: int {1..3}
reason: str
assigned_to: UUID | None
info_deadline, opened_at, closed_at
assign(moderator)
await_information(deadline)
resume(answered | no_response)
appeal(second_moderator)
close(decision | withdrawn)
FR-T-10; state machine C01–C11 (v3.0)

«value»
EvidenceTrail
events: list[LifecycleEvent]
assessment: AssessmentResult | None
condition_reports, statements
attachments (metadata stripped)
gaps: list[str]
FR-T-10, NFR-PR-03 (chat only if disputed)

ModeratorDecision
id, case_id: UUID
decided_by: UUID, decider_role
kind: DecisionKind
reason: str {mandatory}
safeguards: list
resolved_outcome: dict | None
{at_fault, possession}
reverses_decision_id: UUID | None
{set iff APPEAL_OVERTURNED or REVERSAL}
decided_at: datetime
FR-T-11, FR-B-15 (resolved outcome is evidence), ES-11

AppealService
- decisions: DecisionService
- notifier: INotify
file_appeal(case_id, party, reason) {within the appeal window; once per case}
assign_second(case): moderator {not the first decider; Admin if none}
decide_appeal(case_id, moderator, upheld, reason): ModeratorDecision
FR-T-11, UC-I4 4b, C08–C10, sequence (h)

0..* · decided by · «use» · «use» · «use»

CaseStatement
author_id, author_role
BORROWER | LENDER
statement: str
created_at
FR-B-17, UC-B12 6, UC-I4 2 (a moderator question needs schema v2.4)

Notes
• Module 9 changes a lifecycle state only through ILifecycleCommands.apply_decision() (one writer of lifecycle events, A8). A reversal or an overturned appeal never rewinds the loan: ReversalService flags the outcomes that came from the reversed decision through IOutcomeFlags, and the next assessment no longer counts them (ES-11).
• Possession is offered only when the stored previous state allows it (ADR-0009, proposed): ACCEPTED → NOT_HANDED_OVER; ACTIVE or RETURN_INITIATED → WITH_BORROWER, RETURNED or LOST.
• Timer values (response window, appeal window) are configuration, proposed in FR/NFR v1.4.
• The evidence trail is derived from the event log, never stored separately (UC-D6). Search and ranking are Student 1's and Student 2's; Student 3 only logs and evaluates them.

ReversalService
- decisions: DecisionService
- flags: IOutcomeFlags
reverse(decision_id, actor, reason) : ModeratorDecision {REVERSAL}
affected_outcomes(decision): list[UUID] {the loan is never rewound}
FR-T-11, UC-I4 4c, C10, C11, ES-11

uploads (ADR-0014, proposed)

«interface»
StoragePort
put(key, bytes, mime)
signed_url(key, ttl): str
delete(key)
ADR-0014

«use»

AttachmentService
- storage: StoragePort
access(file_owner): Attachment
check(mime, size) {TC-B5}
strip_metadata(bytes): bytes
{EXIF and GPS removed}
url_for(attachment, viewer): str
{parties and moderator only}
NFR-S-06, NFR-PR-01, FR-B-11, FR-B-16

«adapter»
LocalDiskStorage
development and CI

«creates»

«value»
Attachment
transaction_id, uploaded_by
storage_key, content_sha256
mime_type: {png | webp | pdf
metadata_stripped = true
condition_report_id | case_statement_id
attachment row (schema v2.3)

«adapter»
ObjectStorage
deployed (provider chosen in DA-17)

search logging (Module 4)

SearchLogger
log_query(session, query, filters, mode)
log_results(query_id, ranked_ids)
log_click(query_id, item_id)
FR-S-05 (no location or coordinates)

«creates»

«value»
SearchLogEntry
query_id, session (pseudonymous)
query text, filters
mode: HYBRID | KEYWORD_ONLY
shown: list[item_id, rank]
clicked: list[item_id]
at: datetime
FR-S-05; table in DA-23 (Student 1)

«use»

evaluation (Module 10)

ScenarioSuite
- fixtures: list[ScenarioFixture]
run(engine: IAssess): list[ScenarioResult]
load(paths): list[ScenarioFixture]
NFR-Q-01 (13 scenarios by the final, Iteration 1 and 2 fixture files)

«value»
ScenarioResult
scenario_id: str
state_ok, rule_ok: bool
lifecycle_ok, explanation_ok: bool
passed: bool (all four)
NFR-Q-01

1 / 13..*

«value»
ScenarioFixture
scenario_id: str
transaction_context
evidence, steps
expected: ExpectedResult
scenario_fixtures_iteration1.json, …iteration2.json

«value»
ExpectedResult
rule_fired, decision, state
lifecycle_state, hold_reason
explanation facts
NFR-Q-01

1..* · «reads»

SearchEvaluator
- labels: LabelSet
precision_at(k=5), recall_at(k=10)
ndcg_at(k=10)
compare(keyword_only, hybrid)
DA-25; NFR-P-01, P-02 baselines

«value»
LabelSet
queries: 20 to 30, synthetic
judgements: two labellers
agreement(): Cohen's kappa
DA-25

1

interfaces provided by Modules 3 and 5, used by Module 9

«interface»
ILifecycleCommands
apply_decision(transaction_id, decision): event
Student 1, Module 3

«interface»
IEventQuery
load_stream(id)
replay(id)
verify_chain(id)
Student 1, Module 3

«interface»
IOutcomeFlags
flag(outcome_ids, decision_id)
refresh_edges(user_ids)
Student 2, Module 5 (ES-11)

«use»

How to read this diagram
• One diagram for the whole system: every class appears once, in the package of the module that owns it. Packages with a dot are parts of one module (accounts.membership is Module 1, search.retrieval, search.ranking and search logging are Module 4).
• A dependency on a class of another module is shown by its type in the attribute or operation (for example - *notify: INotify*, - *engine: AssuranceEngine*); the module dependencies are the component diagram. Lines cross package borders only where one class uses another directly: Registration creates a User, InvitationService uses the verifier, a BorrowingTransaction holds its open CounterOffer, JobRunner acts through BorrowingService.
• Green text = added or changed in iteration 2; green border = class new in iteration 2. Everything else is as built in iteration 1.
• Interfaces between students: IAssess, IOutcomeSink (Module 5, S2); ILifecycleCommands, IEventQuery, JobRunner, ISearch (Modules 3 and 4, S1); INotify (Module 7, S2); IOutcomeFlags (Module 5, S2); SearchLogger (Module 4, S3).
• Relationship ends: ◆ composition, ◇ aggregation, plain line = association, dashed open arrow = dependency, dashed hollow triangle = realisation. The iteration 1 diagrams, one per student, are in archive_v1/.